localhost:8888/notebooks/Untitled15.ipynb?

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[3]:
```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import LinearSVC
from sklearn.metrics import accuracy_score, classification_report

# 1. Load dataset
df = pd.read_csv("spam.csv", encoding='latin-1')

# Keep only required columns
df = df[['v1', 'v2']]
df.columns = ['label', 'message']

# Convert labels to numbers (ham=0, spam=1)
df['label'] = df['label'].map({'ham': 0, 'spam': 1})

print(df)
```

```
      label                                            message
0         0  Go until jurong point, crazy.. Available only ...
1         0                      Ok lar... Joking wif u oni...
2         1  Free entry in 2 a wkly comp to win FA Cup fina...
3         0  U dun say so early hor... U c already then say...
4         0  Nah I don't think he goes to usf, he lives aro...
...     ...                                                ...
5567      1  This is the 2nd time we have tried 2 contact u...
5568      0              Will Ì_ b going to esplanade fr home?
5569      0  Pity, * was in mood for that. So...any other s...
5570      0  The guy did some bitching but I acted like i'd...
5571      0                         Rofl. Its true to its name

[5572 rows x 2 columns]
```

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[5572 rows x 2 columns]
```

[4]:
```python
# 2. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    df['message'], df['label'], test_size=0.2, random_state=42
)
```

[5]:
```python
# 3. Text vectorization (TF-IDF)
vectorizer = TfidfVectorizer(stop_words='english')

X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)
```

[6]:
```python
# 4. Naive Bayes Model
nb_model = MultinomialNB()
nb_model.fit(X_train_tfidf, y_train)
nb_pred = nb_model.predict(X_test_tfidf)

print("\n===== Naive Bayes Results =====")
print("Accuracy:", accuracy_score(y_test, nb_pred))
print(classification_report(y_test, nb_pred))
```

```
===== Naive Bayes Results =====
Accuracy: 0.9668161434977578
              precision    recall  f1-score   support

           0       0.96      1.00      0.98       965
           1       1.00      0.75      0.86       150

    accuracy                           0.97      1115
   macro avg       0.98      0.88      0.92      1115
weighted avg       0.97      0.97      0.96      1115
```

[8]:
```python
# 6. Prediction function
def predict_email(text):
    text_tfidf = vectorizer.transform([text])
    prediction = svm_model.predict(text_tfidf)
    return "Spam" if prediction[0] == 1 else "Ham"
```

[9]:
```python
# 7. Test examples
print("\n===== Sample Predictions =====")
print(predict_email("Congratulations! You won a free iPhone. Click here now"))
print(predict_email("Hey, are we still meeting tomorrow?"))
```

```
===== Sample Predictions =====
Spam
Ham
```

[ ]: